

Programming Fundamentals Lab (CL1002)

Date: 14/Oct/2024

Course Instructor(s)

Shaheer Ahmad Khan

Lab Mid Exam (B)

Total Time: 90 minutes

Total Marks: 20

Total Questions: 03

Semester: FL-2024

Campus: Karachi

Dept: Software

Engineering (SE)

Submission Instructions:

- You must comment your student ID on top of each file. (Line#1 of your code).
- Name the file for each question according to Roll_No e.g. k24-xxxx_Q1.c, k24-xxxx_Q2.c etc.
- Submission is via a client software so open the application present on the Desktop.
- Enter your username as 24K-xxxx and its assigned password (Default is Fast1234).
- Submission is timed so after the time no submission will be accepted.

Student Name

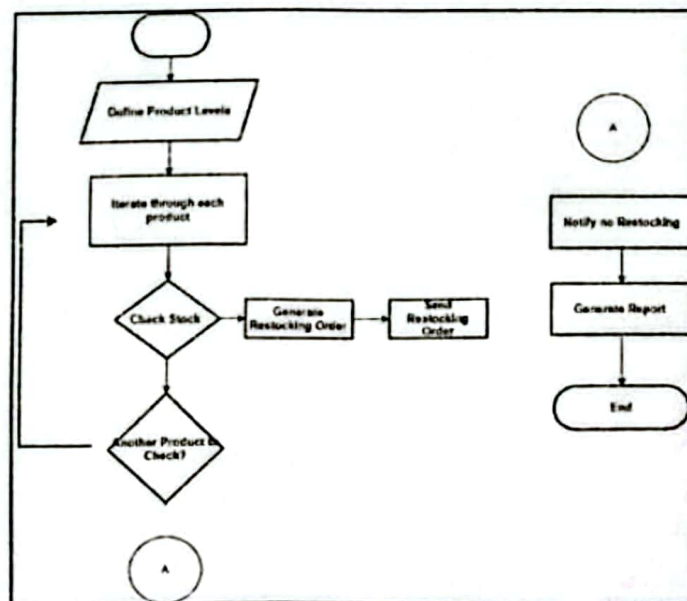
Roll No

Section

Student Signature

CLO # 1: Understand and Analyze flowcharts, PAC (Process-Activity-Control) charts, and IPO (Input-Process-Output) models to represent system workflows and these diagrams into algorithm and pseudocode implementation.

Q1. You are working for a large warehouse that manages various products. Your task is to design an inventory management system that ensures the warehouse operates smoothly by tracking stock levels, alerting the manager when stocks are running low, and automatically generating restocking orders. Based on the provided flowchart, your task is to write pseudocode that mirrors the flowchart. This pseudocode will serve as the blueprint for implementing the actual system. **[5 Marks]**



CLO # 2: Gain hands on experience in writing code that provides the use of logical and bitwise operators to perform efficient data manipulation, and apply decision and nested decision structures in control flow to create dynamic, condition-based logic within C-code

Q2. You are working on a low-level hardware control system that uses bit manipulation to manage device states. You have two integers representing the current state of the device and the target state that you want to achieve. Your task is to write a C program that calculates the minimum number of bit flips required to convert the current state of the device (represented as an integer) to the target state (also represented as an integer) using the bitwise XOR operator. **[7 marks]**

Sample Input & Output:

Enter the current state (as an integer): 10

Enter the target state (as an integer): 20

Minimum number of bit flips required: 4

CLO # 3: Understand and implement code utilize loops for iteration and arrays for data storage and manipulation, demonstrating efficient traversal and management of elements in C-code, while optimizing control flow and enhancing program efficiency.

Q3. You are developing a simple analytics tool for a sports event that tracks the performance of players. The tool needs to determine the second-best performance score from a list of scores submitted by the players. Write a program that prompts the user to enter the total number of scores they want to input. After collecting these scores into an array, your program should traverse the array using loops to find and display the second largest score among the athletes. If there is no second largest score, your program should indicate that as well. **[8 Marks]**

Sample Input & Output:

Enter the total number of scores: 5

Enter the scores:

45 67 89 34 67

The second largest score is: 67